Projecting NBA Rookie-Contract Value using Collegiate Production

A supervised and unsupervised pipeline over 1,042 draftees, 2000–2022

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***Abstract*—How much of an NBA player's production under their rookie contract (four years) can be predicted using only collegiate statistics and high school recruiting rank? Scraping a dataset of the 1042 collegiate players who played at least 15 games of D1 college basketball and went on to play minutes in the NBA between 2000-2022, I attempted to predict their cumulative Value Over Replacement Player (VORP) over the first four years of their career. The initial dataset contained 28 columns: things like points per game (ppg), rebounds per game (rpg), and win shares per 40 minutes (ws/40). I engineered an additional 13 features: position-relative z-scores for relevant counting stats, a volume-weighted three-point percentage metric, and a few interaction terms. Using the resulting dataset, split temporally to prevent data leakage, I trained three regression models (Ridge, XGBoost, and an MLP). The XGBoost architecture was the most successful, with an R^2 on the test set of $.202$, MAE of 1.67 VORP, and an AUC of $.591$ on a simple classification of whether VORP was positive or negative over four years. K-means fit separated by position yielded eight player archetypes that had significant average VORP differences between them, highlighting an area of potential future work.**

I. Introduction

Each June, the eyes of the basketball world turn to the NBA draft, where the culmination of months of work for NBA front offices results in their team selecting a couple of players that they believe will markedly improve the outlook of their franchise. These picks represent a significant financial investment: the #1 overall pick in this year's draft will command a rookie contract with nearly 63 million dollars over four years. With that type of money on the line, it is crucial that these picks are made with an informed process behind them. Scouts watch tape and look at metrics extensively in order to answer: what player in the draft is best suited to improve our chance of winning a championship?

This project asks a scaled-down version of that question: *given only collegiate production and high school recruiting rank, how much of a player's production under their rookie contract are we able to predict?* This project seeks to serve as a precursor to the scouting process, helping to identify high-potential prospects that are worth further investing in, and on the flip side eliminating prospects that are not likely to be valuable NBA contributors.

This problem is first framed as a regression task: attempting to predict the outright cumulative four-year VORP of a given college prospect. This is a more challenging modeling problem, but its target is to identify players that potentially have a high ceiling as a prospect (high projected four-year VORP). A scaled-down version of this problem is simply classifying a player into a positive VORP (above replacement level) or negative VORP (below replacement level) bucket. This is useful in the aforementioned filtering of prospects. If a player has a low probability of achieving at least replacement-level production under their rookie contract, there is little value in continuing to invest resources into scouting them.

I also used K-means for unsupervised clustering to find player archetypes without looking at NBA outcomes. Separating players out by position, there were eight different archetypes identified (3 each for forwards and guards, 2 for centers). This clustering was aimed at exploring how different play styles in college lead to NBA success, even if they were not producing the same raw statistical output as their peers.

II. Related Work

Basketball analytics as a field has a long public history, largely kicked off in the modern era with Dean Oliver's book *Basketball on Paper* [1], which introduced key possession-based efficiency metrics like offensive/defensive rating, as well as features used in this project like true shooting %. A more recent development is our target variable VORP [2]. This metric scales a player's estimated contribution to the game based upon their box-score stats to the number of minutes that they played, giving a singular metric to represent how much value a player provides over the average NBA player while on the court.

The closest peer to this specific project is Kannan et al. [3], who also trained machine learning models on collegiate statistics and biometric data. Their model, however, included draft selection order and was targeted at predicting NBA career longevity. In a similar manner to our results, they reported that collegiate per-possession statistics dominated their predictions. Czasonis et al. [4] is another project that looked at projecting collegiate prospect production in the NBA, but their work was focused around finding players with similar statistical profiles to the prospects in question, then following their NBA career success. This is more similar in scope to the unsupervised learning portion of this project.

III. Data

A. Sources

I scraped two public sources:

- **Basketball-Reference** [14]: per-player college career totals, advanced metrics (WS, WS/40, TOV%, FTr, three-point attempt rate), draft position, and the first four seasons of NBA box-score and advanced metrics.
- **247Sports composite** [15]: high-school recruiting rank, coerced to an integer $1 \ldots 101$, with 101 used as a placeholder for unranked players.

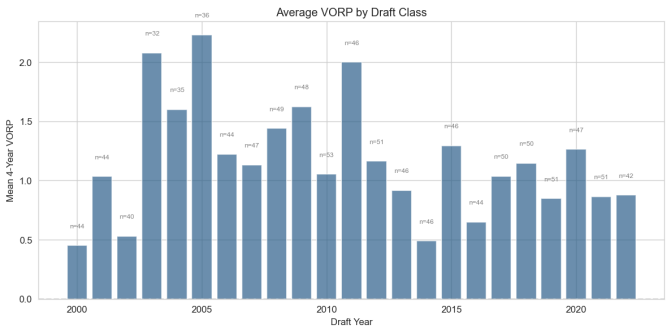

Fig. 1. Distribution of average VORP by draft year

B. Data Cleaning

As this data was collected from the web, one of the most significant hurdles in this project was maintaining data cleanliness. The first challenge was name mismatching. In many cases, the names used when comparing across sources (NBA stat pages, NBA draft information, collegiate statistics, recruiting information, etc.) were different for the same player. This includes cases where suffixes are included or left off (a player may have a Jr. attached to his name in one source and not in another, for example), and there were many cases where shortened/abbreviated names were used across sources (Josh vs. Joshua, for example). This required fuzzy matching to be used to catch many of these errors, as well as some manual mapping between names to be performed in order to make sure all college profiles were properly linked to their NBA production.

There were also a multitude of cases where rows were scraped with missing values, which either required manual input or dropping of the row altogether in some cases.

C. Target

The target variable for both the regression and classification models is `nba_4yr_vorp`, which is the sum of each player's value over replacement player during their first four NBA seasons. For simplicity purposes, these first four years are just the four years after they were drafted. There are some edge cases where players play in the G-League or miss a season as a result of injury during this four-year span, but this is not accounted for during data collection. The distribution of this target variable is heavily centered around 0, which is what you would expect considering that a VORP of 0 indicates an "average" player. However, we do see that the right tail of the distribution is able to extend much further out than the left tail, which is also a sensible result. Star players (players who are accumulating significant VORP) often play more minutes and therefore continue to accumulate more VORP. If a player is playing at a below-replacement level, they would not be expected to continue to get playing time, which limits their ability to continue accumulating negative VORP.

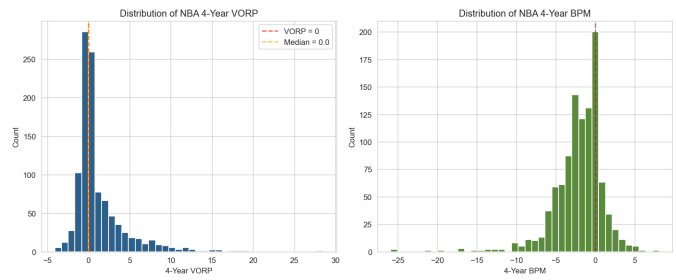

Fig. 2. Distribution of the regression target (4-year VORP) and the complementary BPM target

D. Features

After the extensive data-cleaning process detailed above, I kept 28 raw columns: 10 per-game box-score stats, 8 percentage/advanced stats, 4 win-share columns, 2 physical measurements (height/weight), and 4 categorical indicators (position, college seasons, one-and-done flag, recruit tier). Additionally, I added 13 engineered features, resulting in a final dataset with dimensions 1042×41.

Taking a look at a Pearson correlation matrix of a selected group of features, we see that the strongest raw correlation with four-year VORP is `ws_40`. Raw per-game statistics have very little correlation with the target variable as they mean different things depending on player position, which is what led to some feature engineering decisions.

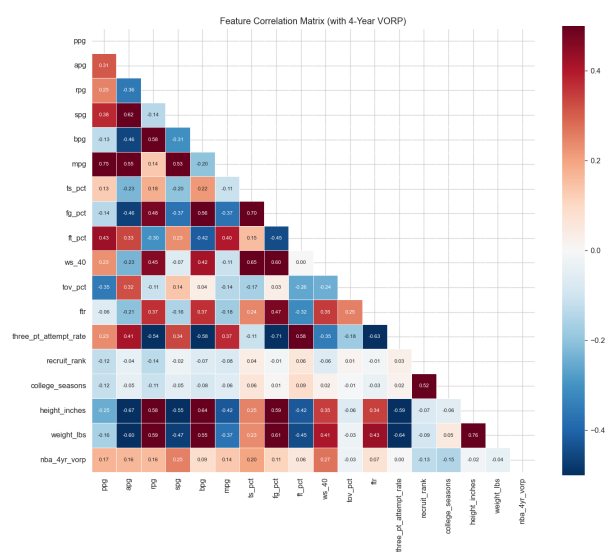

Fig. 3. Pearson correlation matrix for raw features and the regression target. The strongest positive correlate with NBA VORP among pre-engineering features is college WS/40.

IV. Methodology

A. Feature engineering

The biggest issue with our raw data is that per-game statistics are relatively meaningless without per-position context. Four rebounds per game for a guard may be a good statistic, but would be concerning for a center. In order to address this issue, I built *position-relative z-scores* for the primary counting stats in the dataset (PPG, RPG, APG, SPG, BPG, TS%, WS/40), using the per-position means and standard deviations in the training set. The goal of these new features is to encapsulate how much better/worse a player is at certain aspects of the game (rebounding, scoring, etc.) than their positional peers.

I also did a series of feature engineering tests in order to find the most valuable multiplicative features to add to the dataset. There are often non-linear relationships between the value of certain statistics in how they translate to NBA production, which is what these three features attempt to encapsulate:

- `def_versatility` = SPG $\times$ BPG, a single metric that captures defensive disruption, both at the rim and in passing lanes.
- `recruit_x_ws40` = $(101 - \text{recruit_rank}) \cdot$ WS/40, weighting recruiting profile with WS/40 in order to give more context. If a player was perceived to have high potential in high school and then proceeded to produce well in college, this is promising.
- `ws40_per_season` = WS/40 $\div$ college_seasons, weighting production by the number of seasons that a player plays in college. If a freshman and a senior have the same level of production in the season before they declare for the draft, the freshman would typically be considered the more valuable prospect.

Volume-weighted three-point percentage.

Another issue that I encountered in the dataset was that three-point percentage was an exceedingly noisy statistic, especially amongst power forwards and centers. There are many players in the dataset who attempted very few three pointers in their collegiate career, making their raw three-point percentage less useful. If a player shot one three in their career and made it, they are given a 100% three-point percentage, but this player should not be viewed as a "better shooter" than a player who made 40% of their threes on hundreds of attempts. In an attempt to address this issue, I substituted the metric below:

$$\widehat{p}_3 = \frac{p_3 \cdot r_3 + \bar{p}_3 \cdot k}{r_3 + k} \tag{1}$$

where p_3 is raw 3P%, r_3 is three-point attempt rate, $\bar{p}_3 = 0.312$ is the league average across the dataset, and $k = 0.083$ is the 25th percentile of r_3 in the training set. k essentially serves as "fake attempts at the league average," which pulls a player who has an inflated/deflated percentage on low attempt rate towards the league mean. If a player has a large number of attempts throughout their career, the change to their percentage is negligible.

The final engineered feature, `build_score`, is a sum of position-relative height and weight z-scores so that players who are physical outliers at their position are flagged.

B. Train/test split

I chose to split the data temporally at the 2019 draft class, leaving 851 players drafted between 2000-2018 in the training set and 191 players from 2019-2022 in the test set. It was my belief that a random shuffle would leak signal from later players into the model, providing it an unfair advantage. In the hypothetical application of this model, one would not have access to the NBA career outcomes of future prospects, so it would not have been sensible to train the model in this manner.

C. Models

I trained three regressors, each with 5-fold CV inside its own hyperparameter search:

- **Ridge** with $\alpha = 10$, as a simple linear baseline. This model is the simplest, making it ideal for interpretability, as well as exporting to the front end to be used in the demo.
- **XGBoost** with a 50-iteration randomised search over `n_estimators`, `max_depth`, `learning_rate`, `subsample`, `colsample_bytree`, and `min_child_weight`.
- **MLP** (scikit-learn) with 20 search iterations over hidden layer shapes (two to four layers, 16–128 units), initial learning rate, and L^2 regularisation.

Each of these regressors has a matching classifier trained on the binary variable `PosVORP` with the same hyperparameter searching criteria. As mentioned above, this was done to provide a simpler modeling process and to follow the likely real-world application of this model as a precursory sorting tool. All features are standardized with a `StandardScaler` that was fit on the training set.

D. Unsupervised archetypes

For the unsupervised portion of the project, I clustered the full 1042-row scaled feature matrix with K-means. The only further processing done on the data was separating players by position prior to the clustering. This was done to force the K-means to rediscover actual archetypes within the position groupings. Without this separation, the clustering essentially just rediscovered the split between guard, forward, and center, which does not provide much utility. For visualization, I use a global PCA down to two components as a 2D embedding.

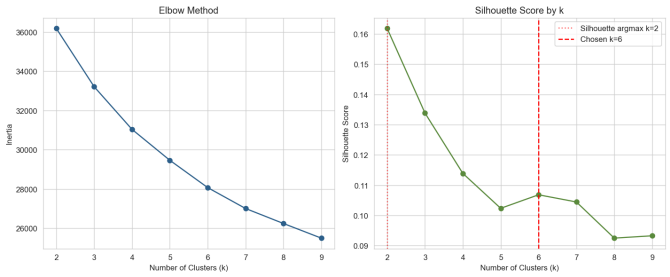

Fig. 4. Silhouette score by k, computed within each position group. Dashed vertical lines mark the chosen k. Guard and forward silhouettes are flat past $k = 3$; centers are too small a group to support more than two stable clusters.

V. Results

A. Regression

Table I shows hold-out performance.

TABLE I
Regression on 4-year VORP, test set (2019–2022, $n = 191$).

Model	R^2	MAE	RMSE
Ridge (baseline)	0.162	1.78	2.35
XGBoost	**0.202**	**1.67**	**2.29**
Neural Network	0.152	1.65	2.36

XGBoost had the most favorable values on R^2 and RMSE. However, the MLP has the lowest MAE, which may be indicative that it had more symmetric errors across the entirety of the training set, but fell victim to a selection of larger misses. The Ridge model is competitive with the other two models considering its simplicity, which is indicative of the efficacy of the feature engineering, as well as the limitations on how much predictive power this feature set has.

B. Classification

On the binary `PosVORP` variable (1 if the player has VORP ≥ 0, 0 if < 0), the XGBoost gets 56.5% accuracy, .491 F1, and .591 AUC-ROC, which is narrowly better across the metrics than the MLP (55.5%, .479, .579). These results reflect the sentiment from the regression: there is a limitation on how much predictive power this feature set has over NBA results.

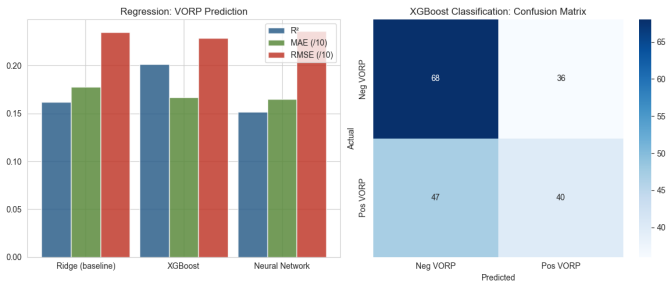

Fig. 5. A comparison of the regression results across the three model architectures, as well as the confusion matrix for the XGBoost classifier

C. Feature importance

Figure 4 shows the mean |SHAP| ranking on the test set. Four of the top five are *position-relative* versions of raw stats, which confirms our initial inference during the feature engineering stage that the raw box-score columns get drowned out by their z-score counterparts. Win Shares per 40 minutes (both position-relative and absolute) sits at the top, followed by playmaking (`apg_vs_pos`) and disruption (`spg`).

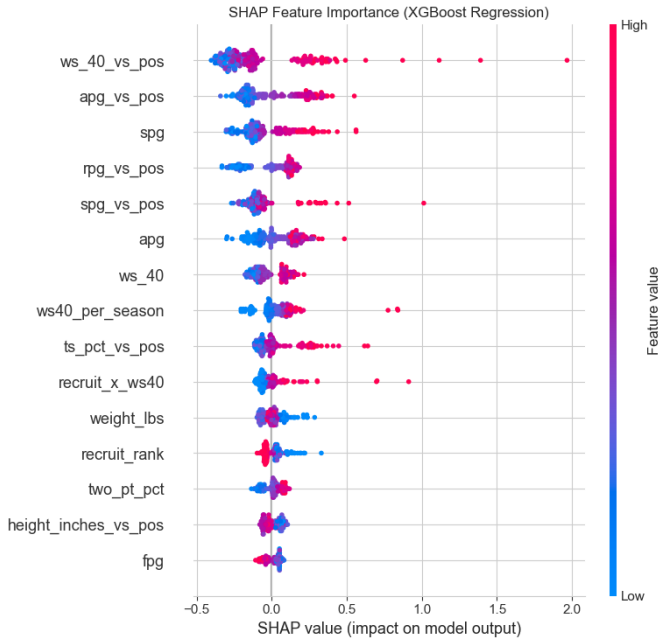

Fig. 6. Mean absolute SHAP values on the test set. Position-relative z-score features occupy four of the top five slots, validating the engineering choice to normalise by position.

D. Archetypes and outcomes

Figure 5 visualizes the PCA embedding of the K-means colored by cluster, as well as the average four-year VORP for each cluster. We can see that the "Scoring forward" archetype has the highest mean VORP at $+2.17$ and the "Traditional center" cluster sits at near-replacement level on average at $+.17$. This significant gap between the mean VORP of these clusters is significant, potentially more significant than the results we were able to obtain from our supervised learning. This suggests that identifying what type of player you are drafting is important, not just how productive they were in their collegiate career.

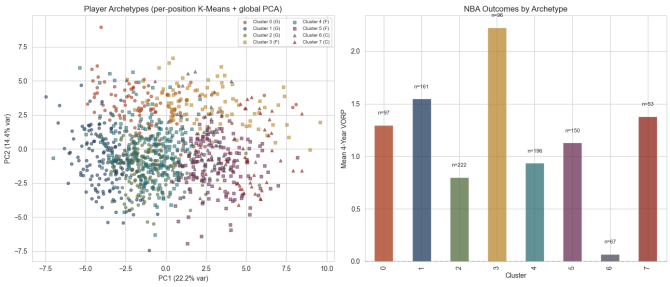

Fig. 7. **Left:** per-position K-means clusters projected onto a shared two-dimensional PCA. Marker shape encodes position, colour encodes cluster. **Right:** mean four-year VORP by cluster, with per-cluster sample size.

E. 2026 projections

When applying the XGBoost regressor to the 2026 draft class, Cameron Boozer of Duke emerged as the top prospect, with a projected four-year VORP of $+9.29$. He was followed by Caleb Wilson, Kingston Flemings, and Allen Graves. Performing these predictions on the 2026 class is also informative in identifying some limitations of the model. Allen Graves is an interesting case in this regard. His counting stats were elite relative to his position in college, which explains why he is so highly viewed when put through the model. However, it is factors that the model does not capture that lead to him being rated lower on actual consensus mock drafts, which are likely more representative of the actual actions that NBA GMs will take come draft day. These factors include strength of schedule in college, as well as physical projections and indicators that may appear during film study that are not apparent when just looking at statistics.

VI. Discussion

The biggest win in feature engineering was the position-relative z-scoring of the primary counting stats. Four of the five most important features in the regression models were these engineered statistics. The Ridge model being comparable to the more sophisticated XGBoost and MLP models is also indicative that, once you remove the position-conditional mean from these raw statistics, much of the predictive power of this feature set is linear. The shrinkage estimator for three-point percentage did little to make this feature more useful in the predictions themselves, but the more significant impact was removing the 100% and 0% outliers in the dataset. These values were not indicative of true shooting quality and only served to disturb the accuracy of the model.

The per-position K-means gives up some global utility as compared to clustering across all players in the dataset, but this choice allows us to discover novel archetypes within the dataset rather than just rediscovering the boundaries between positions, which is something that we already have in the dataset. This clustering also provides utility in predicting NBA career outcomes. The spread between the lowest average (.17) and highest average (2.17) mean VORP is not insignificant and is indicative of how archetyping players can carry value even without looking at NBA outcomes during the clustering process.

The MLP lost to XGBoost on both RMSE and R^2 despite getting the same hyperparameter budget, which is what you'd expect with only 851 training rows. Tree boosting picks up most of the available signal from greedy axis-aligned splits, and you usually need 10–100x more data before a neural net's learned interactions actually pay off.

I'd read the 0.202 hold-out R^2 as an upper bound on what college box-score plus recruiting-rank features can deliver, *not* as a ceiling on draft prediction itself. There's a lot of variance the features just can't see: strength-of-schedule and pace-adjusted ratings, shot-location splits, anthropometric and combine numbers (publicly logged but with $> 30\,\%$ missingness before 2014), and injury/developmental trajectory. Era drift is another thing to consider here. The average three-point attempt rate for the draft classes nearly doubled across the training set, which represents a drift in playstyle that is not captured by the model. An ideal prospect in 2000 is not the same as an ideal prospect in 2022, which is a concept not well captured by our methods.

VII. Conclusion and Future Work

Using only college box scores and high-school recruiting rank, a tuned XGBoost model explains about $20\,\%$ of the variance in four-year NBA VORP on a 2019–2022 hold-out. Unsupervised clustering within each position group finds eight interpretable archetypes whose NBA outcomes differ by roughly 10x, which is a nice reminder that "what kind of prospect" is an important signal before you even get to "how good" a prospect is.

Future work: pull in strength-of-schedule adjustments, combine measurements, and maybe per-game shot-chart data (which the NCAA is starting to make available publicly). It would also be worth ensembling the model with NBA-side scouting priors like mock-draft consensus to see if there's any predictive content left after the college stats. Finally, the whole pipeline could be retargeted to predict *second-contract* value instead of rookie-contract value, which would do a better job capturing long-term NBA impact. I am also interested in doing further exploration of how unsupervised clustering can be used in predicting NBA early-career outcomes. Early results from this project seem to be promising, so I would be interested to explore how combining this with other supervised methods might yield stronger results.

VIII. Individual Contributions

This is a solo project. All data acquisition, cleaning, feature engineering, modelling, frontend implementation, and writing was done by Colin Hanley.

IX. Personal Reflection

My most significant takeaway from this project was on the data-cleaning side of things, as I had mostly worked with pre-cleaned data in the past. This was also the most challenging part of the project, as I was forced to spend time manually checking some of the data and fixing entries by hand in some cases. I would allocate more time to the exploration of the

unsupervised methods in this project if I were to go back and do anything differently.

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Appendix A
AI Usage Log

The following log records uses of Claude (Anthropic) during this project in accordance with the course's AI/LLM policy. Each entry names the phase, summarizes the prompt, summarizes Claude's output, and documents what was accepted, modified, or rejected.

```
--------------------------------------------
AI Log Entry #1
--------------------------------------------
Date:    2026-04-17
Tool:    Claude Opus 4.7 (1M context)
Used by: Colin Hanley
Phase:   Feature Engineering

PURPOSE:
Replace raw 3P% with a volume-weighted variant to kill
noise from low-attempt 0%/100% players.

PROMPT:
"for 3P%, I want to make it volume weighted
(w attempt rate) instead of raw percentage,
since there are players with 100% or 0% on
rly low attempt"

AI OUTPUT (Summary):
Proposed two approaches: (1) simple product
3P% x attempt_rate, or (2) shrinkage toward
league mean with k = 25th-pct attempt rate.
Recommended (2) for preserving % interpretation.

YOUR ACTION:
Adopted (2). Verified formula in Python,
checked that extreme raw values shrink
sensibly (Nesmith 82.5% stays high, 100%-
on-1-attempt players shrink to league mean).

FINAL RESULT:
Implemented in src/preprocess.py engineer_features
as `three_pt_pct_vw`. Replaces raw column in
feature matrix. Correlation with VORP improved
from +0.023 to +0.033.
--------------------------------------------
```